Project Solution Report: Zomato Data Analysis

1. Project Overview:

Title: Zomato Data Analysis & Interactive Dashboard.

Tools Used: Excel, SQL (SQLite), Power BI.

2. Project Objective:

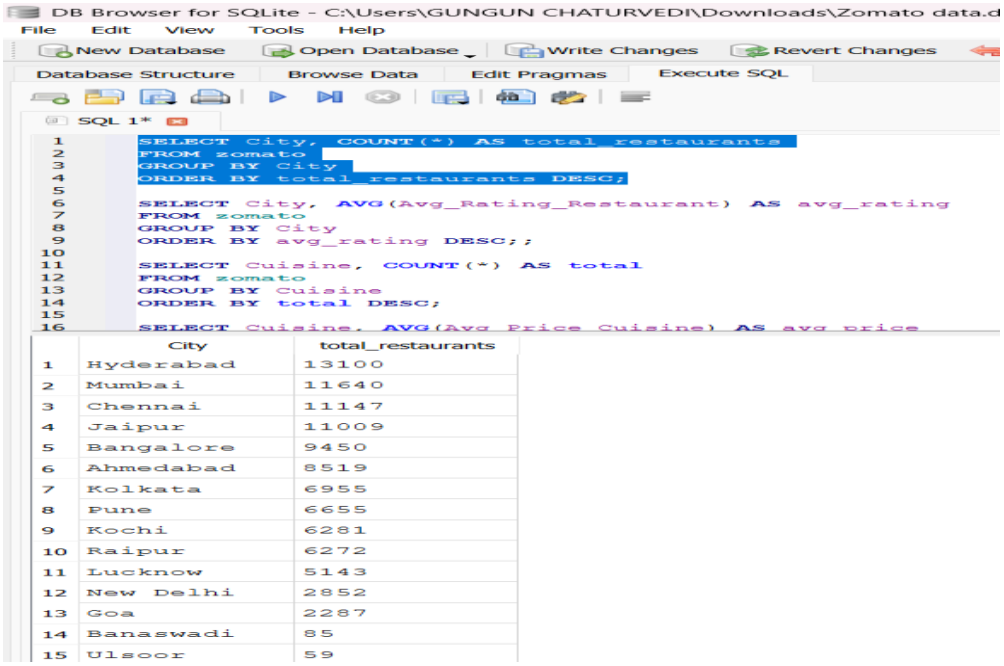
The primary objective of this project is to perform a comprehensive analysis of Zomato's restaurant data to uncover actionable business insights. By leveraging a multi-tool approach (Excel, SQL, and Power BI), the project aims to identify market trends, restaurant density across different cities, customer preferences for various cuisines, and the relationship between pricing and ratings. The ultimate goal is to transform raw data into a visual story that can help in strategic decision-making for the food-tech industry.

3. Key Business Questions & Solutions:

A. City-wise Distribution:

--Query 1: To find the total number of restaurants in each city

```
SELECT City, COUNT(*) AS total_restaurants
FROM zomato
GROUP BY City
ORDER BY total_restaurants DESC;
```

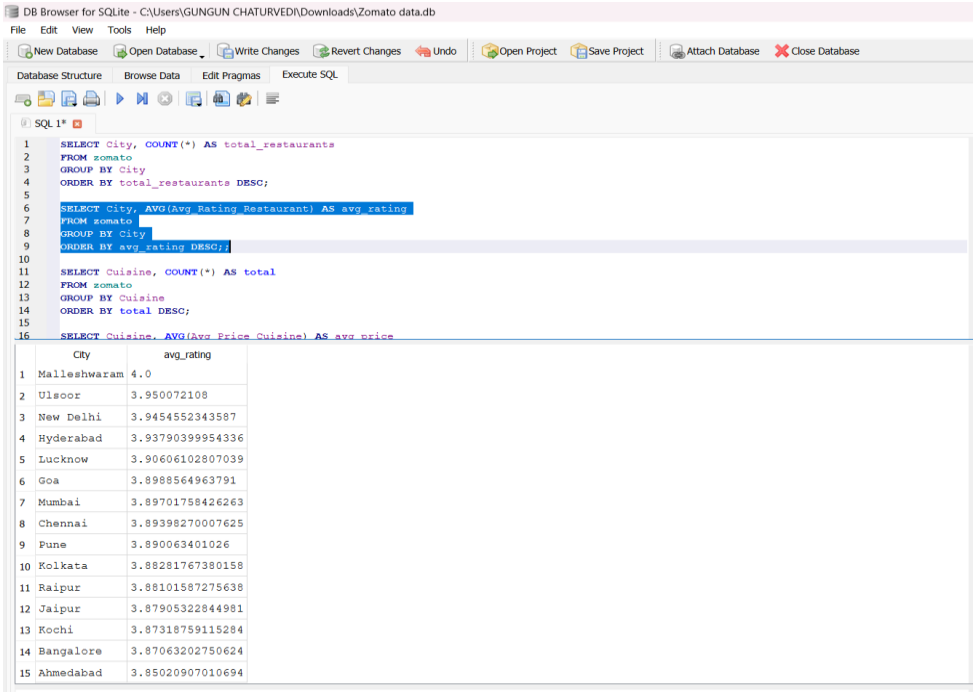


	City	total_restaurants
1	Hyderabad	13100
2	Mumbai	11640
3	Chennai	11147
4	Jaipur	11009
5	Bangalore	9450
6	Ahmedabad	8519
7	Kolkata	6955
8	Pune	6655
9	Kochi	6281
10	Raipur	6272
11	Lucknow	5143
12	New Delhi	2852
13	Goa	2287
14	Banaswadi	85
15	Ulsoor	59

B. Rating & Quality Analysis:

-- Query 2: Average Rating by City

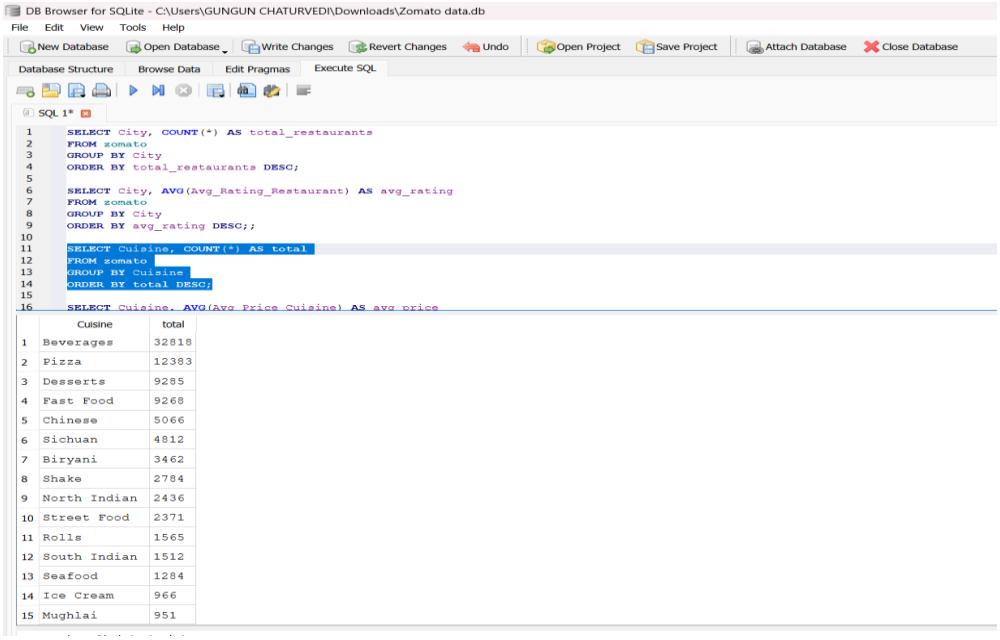
```
SELECT City, AVG(Avg_Rating_Restaurant) AS
avg_rating
FROM zomato
GROUP BY City
ORDER BY avg_rating DESC;;
```



	City	avg_rating
1	Malleshwaram	4.0
2	Ulsoor	3.950072108
3	New Delhi	3.9454552343587
4	Hyderabad	3.93790399954336
5	Lucknow	3.90606102807039
6	Goa	3.8988564963791
7	Mumbai	3.89701758426263
8	Chennai	3.89398270007625
9	Pune	3.890063401026
10	Kolkata	3.88281767380158
11	Raipur	3.88101587275638
12	Jaipur	3.87905322844981
13	Kochi	3.87318759115284
14	Bangalore	3.87063202750624
15	Ahmedabad	3.85020907010694

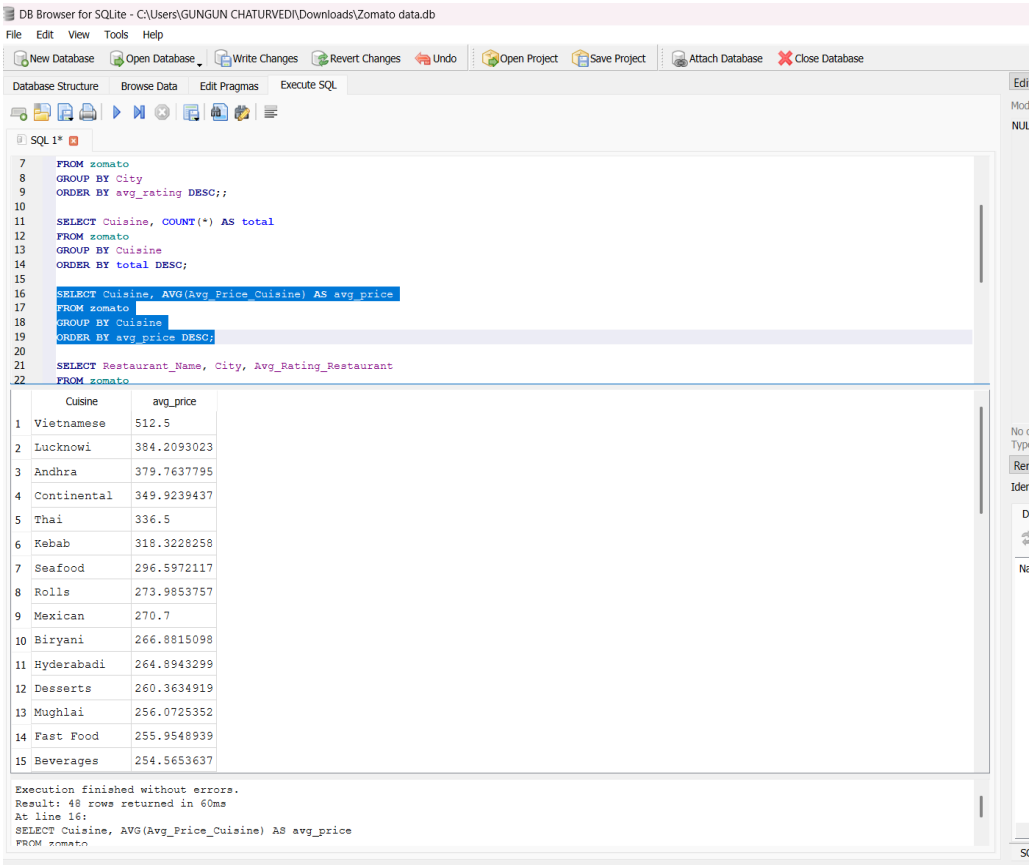
-- Query 3: Most Popular Cuisines

```
SELECT Cuisine, COUNT(*) AS total
FROM zomato
GROUP BY Cuisine
ORDER BY total DESC;
```



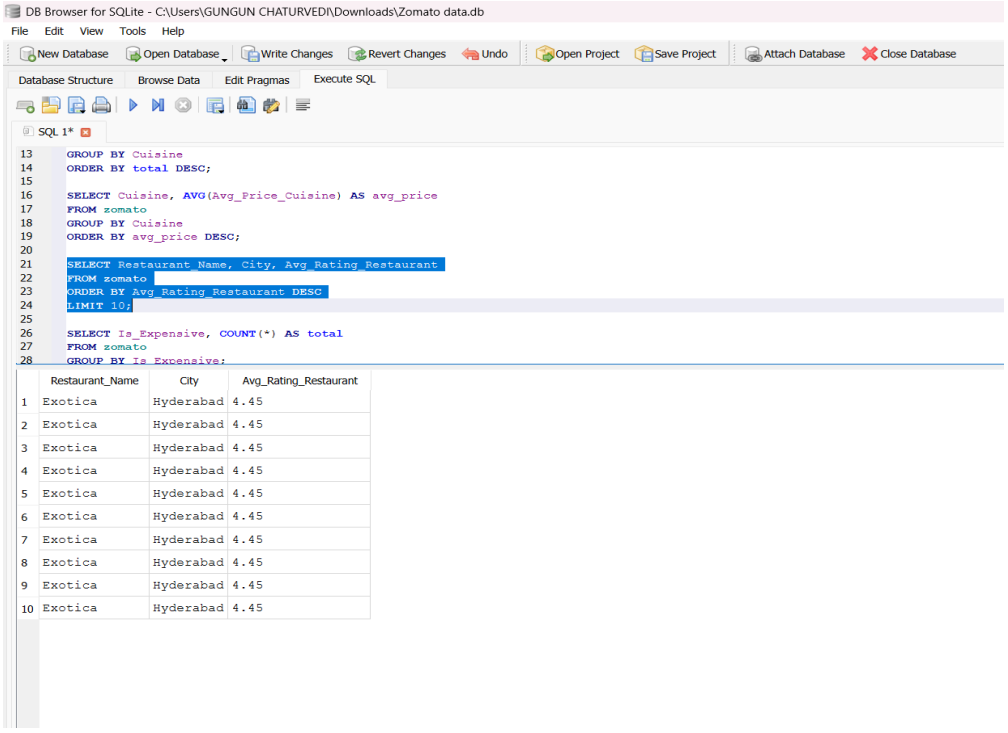
--Query 4: Average Price by Cuisine

```
SELECT Cuisine, AVG(Avg_Price_Cuisine) AS avg_price
FROM zomato
GROUP BY Cuisine
ORDER BY avg_price DESC;
```



--Query 5: Top 10 Restaurants by Rating

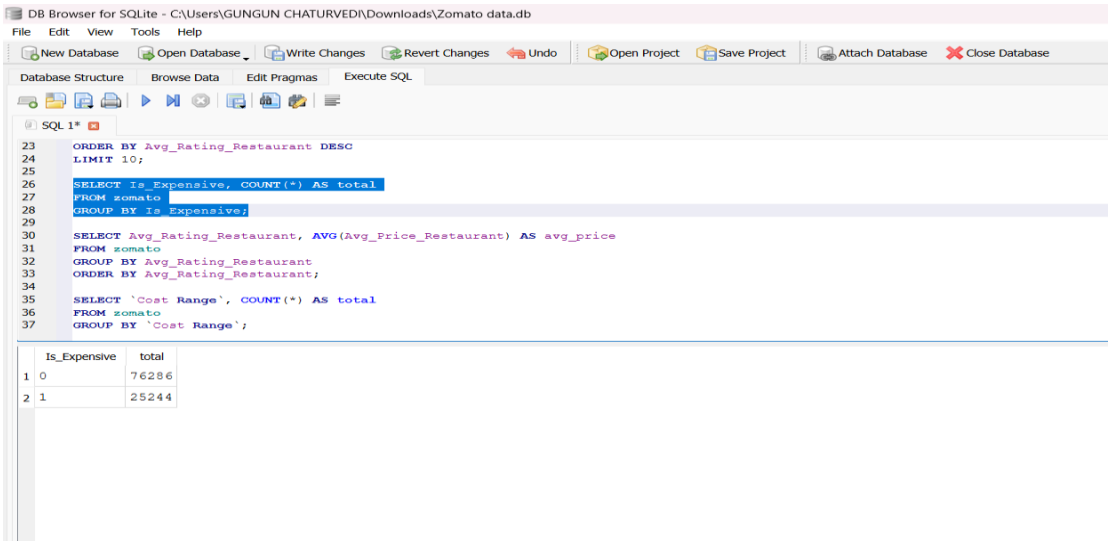
```
SELECT Restaurant_Name, City, Avg_Rating_Restaurant
FROM zomato
ORDER BY Avg_Rating_Restaurant DESC
LIMIT 10;
```



	Restaurant_Name	City	Avg_Rating_Restaurant
1	Exotica	Hyderabad	4.45
2	Exotica	Hyderabad	4.45
3	Exotica	Hyderabad	4.45
4	Exotica	Hyderabad	4.45
5	Exotica	Hyderabad	4.45
6	Exotica	Hyderabad	4.45
7	Exotica	Hyderabad	4.45
8	Exotica	Hyderabad	4.45
9	Exotica	Hyderabad	4.45
10	Exotica	Hyderabad	4.45

--Query 6: Expensive vs Affordable Restaurants

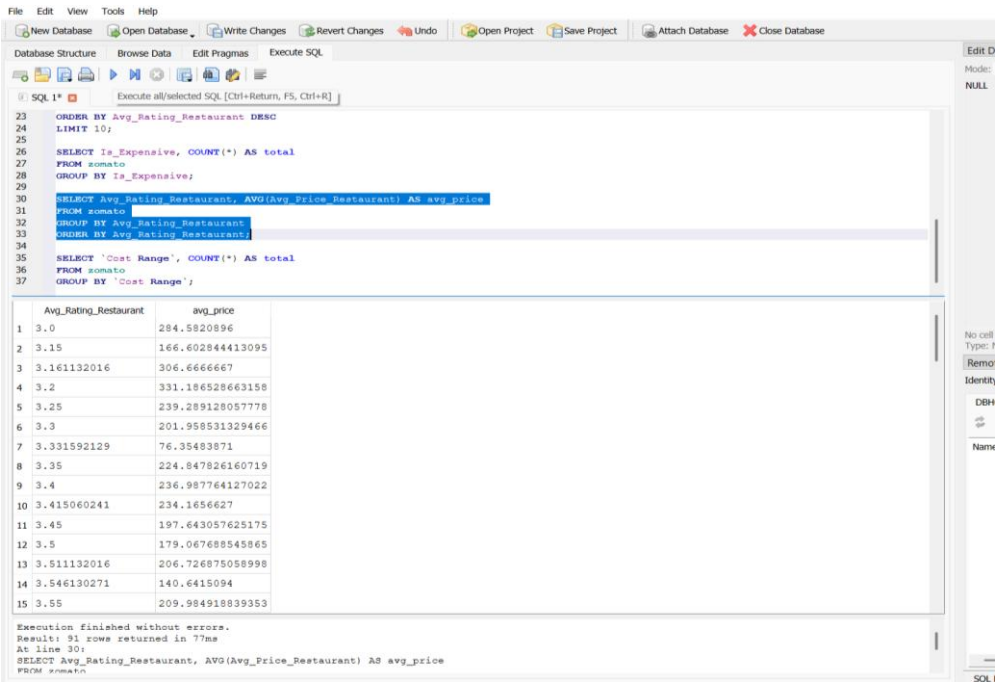
```
SELECT Is_Expensive, COUNT(*) AS total
FROM zomato
GROUP BY Is_Expensive;
```



	Is_Expensive	total
1	0	76286
2	1	25244

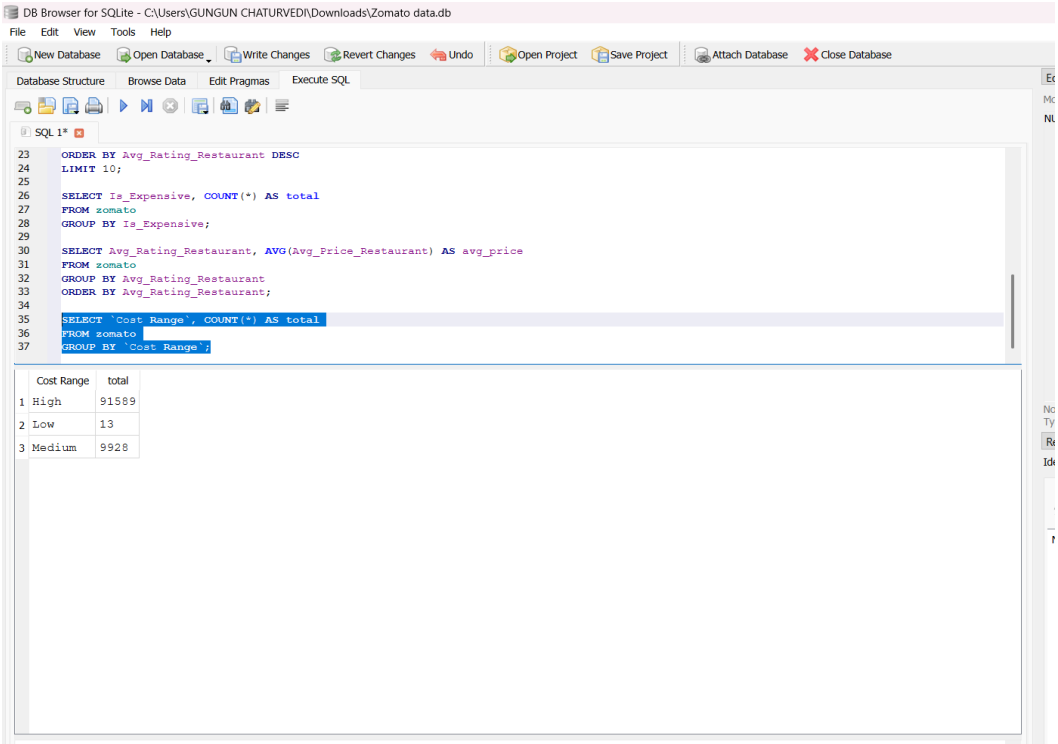
--Query 7: Price vs Rating Analysis

```
SELECT Avg_Rating_Restaurant,
AVG(Avg_Price_Restaurant) AS avg_price
FROM zomato
GROUP BY Avg_Rating_Restaurant
ORDER BY Avg_Rating_Restaurant;
```



--Query 8: Cost Range Distribution

```
SELECT `Cost Range`, COUNT(*) AS total
FROM zomato
GROUP BY `Cost Range`;
```



Conclusion: This project successfully demonstrates a complete data analysis workflow—from raw data cleaning to extracting meaningful business intelligence. By using **SQL (SQLite)**, I was able to handle a large dataset of over 50,000+ rows and extract specific answers that are often difficult to identify in a standard spreadsheet.

Key Learnings from this project:

- **Data Integrity:** Identified that while quantity is high in cities like Hyderabad, quality (ratings) varies significantly across different regions.
- **Market Segmentation:** Segmented restaurants into 'Expensive' and 'Affordable' categories to understand the pricing strategy of various cuisines.